# AI Study Assistant - Technical Documentation

## 1. Project Overview

The **AI Study Assistant** is a web-based application developed to demonstrate the practical use of generative artificial intelligence in education. It helps students understand academic topics, summarize information, and prepare study materials through an interactive interface. The application connects to the OpenAI API, allowing a large language model (LLM) to generate responses to student requests.

## 2. Technologies Used

The application uses **HTML** for page structure, **CSS** for visual presentation, and **PHP** for server-side processing. PHP communicates with the OpenAI API and passes generated results back to the webpage. Development and testing take place locally on Ubuntu using the PHP installation provided by XAMPP.

## 3. How the Application Works

The application follows a simple **client-server workflow**. First, a student enters a question or requests a study task in the web interface. The browser submits that input to the PHP backend, which prepares a request for the OpenAI API. The request includes the user's text and uses an API key for authentication.

OpenAI's language model processes the request and returns generated text. The PHP backend reads the API response and displays the result in the browser. Depending on the request, the assistant can **explain difficult concepts**, **summarize passages**, or **generate practice questions and quizzes**. Responses are generated dynamically rather than selected from a fixed set of predefined answers.

## 4. Execution and Requirements

The project requires PHP, an internet connection, and a valid OpenAI API key. From the project directory, the local development server can be started with:

```
/opt/lampp/bin/php -S localhost:8000
```

The application is then available in a browser at **http://localhost:8000**.

## 5. Conclusion

This project demonstrates the integration of a generative AI service into a conventional web application. It illustrates how a frontend, PHP backend, and external LLM API work together to provide interactive educational assistance.